Lorenzo Valente

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INSPIRE-HEP | ORCID | GitHub | Hugging Face | LinkedIn | Website

Education

By March 2027	**PhD in Physics** University of Hamburg, Germany **Research Focus**: Machine Learning for Fast Calorimeter Simulation, in collaboration with DESY. Supervisor: Prof. Gregor Kasieczka, co-supervisor Dr. Frank Gaede
March 2023	**Master in Nuclear and Subnuclear Physics** University of Bologna, Italy Thesis title: "A variational autoencoder application for real-time anomaly detection at CMS". Supervisor: Prof. Daniele Bonacorsi
April – August 2021	**Erasmus+ Exchange Semester** Ludwig Maximilian University of Munich, Germany Courses: Machine Learning in Fundamental Physics (S. Krippendorf), Radiation Detectors for Medical Applications.
July 2020	**Bachelor in Physics** University of Bologna, Italy Thesis title: "Reconstruction of non-prompt charmed baryon Λ_c with Boosted Decision Trees technique" (ALICE experiment). Supervisor: Prof. Andrea Alici
October 2018	**Master in Classical Piano** Conservatory of Music Luisa D'Annunzio, Pescara, Italy

Talks

Conference talks

- **8th Inter-Experimental LHC Machine Learning Workshop** — **CERN, Switzerland**
  "Transferable Shower Simulation across Calorimeter Designs" — 11 November 2026
- **European AI for Fundamental Physics Conference (EuCAIFCon)** — **Heidelberg, Germany**
  "Cross-Geometry Shower Simulation via Pre-training and Transfer" — 24 August 2026
- **14th Large Hadron Collider Physics Conference (LHCP)** — **Paris, France**
  "ML and Detector Simulation" (invited) — 21 May 2026
- **23rd International Workshop on Advanced Computing and Analysis Techniques in Physics Research (ACAT)** — **Hamburg, Germany**
  "Cross-Geometry Fast Electromagnetic Shower Simulation" — 11 September 2025
- **7th Inter-Experimental LHC Machine Learning Workshop** — **CERN, Switzerland**
  "Cross-Geometry Fast Electromagnetic Shower Simulation" — 20 May 2025

Other talks

- **Glühwein Workshop 2025** — **Aachen, Germany**
  "Cross-Geometry Fast Electromagnetic Shower Simulation" — 15 December 2025
- **8th Round Table on Deep Learning at DESY** — **Hamburg, Germany**
  "Cross-Geometry Transfer Learning in Fast Electromagnetic Shower Simulation" — 14 November 2025
- **DPG Spring Meeting 2025** — **Göttingen, Germany**
  "Adaptive Generative Modeling for Accelerated Calorimeter Simulations via Domain Transfer" — 3 April 2025

- **DPG Spring Meeting 2024** — **Karlsruhe, Germany**
  "Adaptive Generative Modeling for High-Granularity Calorimeters" — 8 March 2024
- **ML-INFN Weekly Meeting** — **INFN, Italy**
  "Joint Variational Auto-Encoder for Anomaly Detection in High Energy Physics" — 17 July 2023

Research Experience

October 2023 – present **Doctoral Researcher**
University of Hamburg, Germany
Machine learning for fast calorimeter simulation, based on generative models and on transfer learning to unseen detector geometries.
Supervisors: Prof. Gregor Kasieczka and Dr. Frank Gaede, in collaboration with Dr. Katja Krüger

April – July 2023 **Summer Student**
Instituto de Astrofísica de Canarias (IAC), Spain
Deep learning classification of heavy nuclei in cosmic rays (iron spectrum), ensemble approach reaching AUC 0.90.
Funded by an Erasmus+ traineeship grant.
Supervisors: Dr. Mónica Vázquez Acosta and Dr. Miguel Molero González

June 2022 – March 2023 **Master's Thesis Research**
University of Bologna, Italy
Variational autoencoder for real-time anomaly detection at the CMS Level 1 trigger on public simulation, quantisation-aware training with QKeras and hls4ml.
Supervisor: Prof. Daniele Bonacorsi

Teaching

AI in Physics, MSc course, University of Hamburg, tutor, 2025

Python tutorial, BSc course, University of Hamburg, tutor, 2025

BSc thesis co-supervision, Investigating the Behavior of Generative Models in Particle Physics under Recursive Training, University of Hamburg, 2025

Schools and Workshops

ELLIS Winter School on Foundation Models, Amsterdam, Netherlands — March 2025

Generative Modeling Summer School, Eindhoven, Netherlands — July 2024

Machine Learning with FPGAs for Particle Physics, Bologna, Italy — November 2022

Quantum Sensing, Information Processing and Computing, Bologna, Italy — July 2022

Technical Skills

Simulation & HEP Software: ROOT, Geant4, DD4hep (ddsim, k4geo, ddfastsim), Key4hep, LCIO, uproot, awkward

Programming Languages: Python, C/C++, Bash, LabVIEW

Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, ONNX export, mixed-precision training, experiment tracking (Comet)

Computing Infrastructure: Slurm (DESY Maxwell), HTCondor (DESY NAF), Singularity containers, distributed multi-GPU training, HDF5

Real-Time & FPGA: hls4ml, QKeras, Vivado HLS, VHDL

Software Engineering: Git, CI, pytest, Docker

Languages

Italian (Native), English (Fluent), Spanish (B2), German (B1)

Lorenzo Valente

Publications

Publications

* *corresponding author*

Journal articles and preprints

- F. Gaede, G. Kasieczka, **L. Valente***, *Cross-Geometry Transfer Learning in Fast Electromagnetic Shower Simulation*, JINST **21** (2026) P07037.
  Contribution: conceived and implemented the transfer framework, ran all trainings and analysis, wrote the paper.
- H. Day-Hall*, ..., **L. Valente**, *CaloClouds3: Ultra-fast Geometry-Independent Highly-Granular Calorimeter Simulation*, JINST **21** (2026) P03018.
  Contribution: trained and tuned successive model versions, fed improvements from my cross-geometry transfer work back into the model, contributed figures and manuscript review.
- T. Buss, ..., **L. Valente***, *Transferable Fast Calorimeter Shower Generation via Multi-Geometry Pre-training*, JINST **21** (2026) P10041, alphabetical authors.
  Contribution: led the study design, dataset production, training campaigns and analysis, wrote the paper.

Conference proceedings

- T. Buss, ..., **L. Valente***, *Point-Cloud Generative Models for Fast Calorimeter Simulation across Particles and Geometries*, arXiv:2609.30403 (2026), LHCP 2026, accepted in PoS.
  Contribution: gave the invited talk and wrote the paper.
- T. Buss, ..., **L. Valente***, D. Weber, *Cross-Geometry Transfer and Model Collapse in Point Cloud Calorimeter Shower Generation*, arXiv:2609.28661 (2026), ACAT 2025, submitted to PoS.
  Contribution: gave the talk and wrote the paper, the model collapse study comes from the BSc thesis I co-supervised.
- T. Buss, ..., **L. Valente**, *Jevons' Paradox and Fast Generative Simulation for HEP: Why Realistic Benchmarking is Essential*, arXiv:2609.24306 (2026), SciPost Phys. Proc. (submitted).
  Contribution: co-author through CaloClouds3, the model the article builds on, which I trained and tuned across its versions.
- **L. Valente*** et al., *Joint Variational Auto-Encoder for Anomaly Detection in High Energy Physics*, PoS **ISGC&HEPiX2023** (2023) 014.
  Contribution: developed the model, ran the trainings and the analysis and wrote the paper, MSc work on real-time anomaly detection at CMS.